

Re: harmonics on the three-sphere — how the “honest” matrices were built

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1 The two comparisons, side by side

Your own check computes, for a candidate abstract matrix T'_i and the combined column of basis functions $v = (\psi_{1,1}; \psi_{1,0}; \psi_{1,-1})$, the ordinary matrix–vector product $T'_i \cdot v$, and compares it entrywise to $(\hat{L}_i \psi_{1,1}; \hat{L}_i \psi_{1,0}; \hat{L}_i \psi_{1,-1})$.

That is a well-defined thing to compute, but it is a *different* operation from asking: “what matrix, in the basis $\{\psi_{1,1}, \psi_{1,0}, \psi_{1,-1}\}$, actually represents $\hat{L}_i$?”

The second question is answered like this: apply $\hat{L}_i$ to *one basis function at a time*, and expand the result back in the same basis:

$$\hat{L}_i \psi_{1,m} = \sum_{m'} M_{m'm} \psi_{1,m'},$$

so that column m of the matrix M is exactly the (unique) set of coefficients needed to write $\hat{L}_i \psi_{1,m}$ back in the basis. This M is the honest matrix representation of $\hat{L}_i$.

If $(T'_i \cdot v)_{\text{row}}$ happens to equal $\hat{L}_i \psi_{1,\text{row}}$ for every row, that tells you T'_i and M are related by a **transpose**: $(T'_i \cdot v)_{\text{row}} = \sum_{\text{col}} (T'_i)_{\text{row,col}} \psi_{1,\text{col}}$ uses the *rows* of T'_i , whereas the honest construction of M above reads off the *columns*. For a diagonal or otherwise symmetric matrix the row and column pictures coincide, which is exactly why your $\hat{L}_z = T'_z$ check gave a clean match — $T'_z = \text{diag}(1, 0, -1)$ is trivially symmetric. It is not a coincidence that the one clean component was the diagonal one.

2 Building M three independent ways

I built the honest matrix M_x (and M_y, M_z) three separate ways, to make sure the conclusion did not depend on which technique was used.

Method A — direct solve. Apply $\hat{L}_x$ to each of $\psi_{1,1} = \sin \theta e^{i\phi}$, $\psi_{1,0} = \cos \theta$, $\psi_{1,-1} = \sin \theta e^{-i\phi}$ in turn, then solve the linear system for the coefficients that re-express each result in the same basis (matching the $e^{i\phi}$, constant, and $e^{-i\phi}$ parts separately).

Method B — coefficient extraction. Same idea, but the coefficients are extracted by evaluating the residual at a few sample values of ϕ and solving directly, rather than a general symbolic solve — a cheap cross-check on Method A.

Method C — inner-product (bra–ket) projection. The cleanest of the three, and the one I would recommend as the standard going forward:

$$M_{m'm} = \frac{\langle \psi_{1,m'} | \hat{L}_i | \psi_{1,m} \rangle}{\langle \psi_{1,m'} | \psi_{1,m'} \rangle}, \quad \langle f | g \rangle := \int_0^\pi \int_0^{2\pi} \bar{f} g \sin \theta d\phi d\theta.$$

This is exactly the ordinary quantum-mechanical way of reading off a matrix element, and it sidesteps the row/column ambiguity entirely, since each entry of M is defined by its own independent integral.

All three methods gave, for your specific choice $\chi_{1,1} = \chi_{1,-1} = \sin \theta$, $\chi_{1,0} = \cos \theta$, exactly:

$$M_x = \begin{pmatrix} 0 & -\frac{1}{2} & 0 \\ -1 & 0 & 1 \\ 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad M_y = \begin{pmatrix} 0 & \frac{i}{2} & 0 \\ -i & 0 & -i \\ 0 & \frac{i}{2} & 0 \end{pmatrix}, \quad M_z = \text{diag}(1, 0, -1).$$

Comparing these directly to your own T'_x, T'_y, T'_z (equations 19–21 of your note):

$$M_x = -(T'_x)^\top, \quad M_y = +(T'_y)^\top, \quad M_z = T'_z,$$

each an exact identity (zero residual), independent of which of the three methods above produced M .

3 Why y and z looked clean and x did not

T'_z is diagonal, hence symmetric, so transpose is invisible there. For T'_y , the specific entries happen to satisfy $(T'_i \cdot v)_{\text{row}} = M_{\text{row}, v}$ numerically for this particular v — a coincidence of these specific matrix entries, not a general fact about T'_y (it is not itself symmetric or antisymmetric). T'_x 's entries do not share that coincidence, which is exactly where the apparent -1 appears. In short: your arithmetic in equations (1)–(29) is correct throughout — I rechecked every one of them independently and found no error — the -1 has this precise, structural origin rather than being a mistake anywhere in the calculation.

4 One small, unrelated thing, for completeness

Separately: in your equation (18), S^{-1} has a sign slip in one entry (the $(2, 3)$ position). It did not affect anything downstream — your own T'_x, T'_y, T'_z in equations (19)–(21) are exactly what you get from the *correct* inverse, so nothing needs to be revisited there.

5 Looking forward to your reasoning

I'd be very interested to see why you were expecting $\hat{L}_i \psi = T'_i \psi$ to hold identically for all three generators — if there's a different, perhaps more physically natural convention where that holds exactly (rather than up to the transpose above), that would be worth understanding properly before we go further with the three-sphere case.